Freshness Classification of Korean Food

Digital Image Processing

2026

Contents

1. Datasets

1.1. Available Datasets

Seven publicly available datasets were identified and surveyed. Table 1 summarizes their properties. Together they cover fruits, vegetables, and citrus varieties, with fresh/rotten binary labels and, in one case, a three-class annotation including formalin-mixed samples. One dataset (Ripening Stages) uses an object detection format (YOLOv5) rather than image classification and is therefore excluded from training experiments.

Table 1: Datasets surveyed. †Object detection format — excluded from classification experiments.

Dataset	Categories	Classes	Images	Source
FruitVision	Apple, Banana, Grape, Mango, Orange	Fresh / Formalin / Rotten	10,154	Mendeley
Lemon Varieties	7 lemon varieties	Fresh / Rotten	1,956	Mendeley
Ripening Stages†	Strawberry, Avocado	Unripe / Partial / Ripe / Rotten	—	Mendeley
Fruits Classification	Peach, Pomegranate, Strawberry	Fresh / Rotten	1,655	Mendeley
Fruits Quality	12 mixed fruits	Fresh / Rotten	359	Kaggle
Fruits Fresh/Rotten	Apple, Banana, Orange	Fresh / Rotten	13,599	Kaggle
Fresh & Stale	Apple, Banana, Cucumber, Okra, Orange, Potato, Tomato	Fresh / Rotten	27,317	Kaggle
Total (classification datasets)			**55,040**	

None of the available datasets contains Korean food items. The FruitVision dataset is the most challenging due to its three-class structure, where formalin-mixed samples are visually similar to fresh ones — making it a useful proxy for detecting subtle spoilage indicators, analogous to the difficulty expected in fermented Korean food.

1.2. Korean Food Dataset — Proposal

No labeled dataset for Korean food freshness currently exists in public repositories. Table 2 lists the target food categories and classification states for the proposed dataset.

Table 2: Target categories for the proposed Korean food freshness dataset.

Category	Romanization	States	Notes
Kimchi	kimchi	Fresh / Rotten	Color and mold during spoilage
Rice cake	tteok	Fresh / Rotten	Mold visible on surface
Side dishes	banchan	Fresh / Rotten	Vegetable-based, wilting
Cooked rice	bap	Fresh / Rotten	Mold indicator
Korean apple	sagwa	Fresh / Rotten	Similar to Western apple
Mandarin (Jeju)	gyul	Fresh / Semifresh / Rotten	Jeju variety, citrus
Shine muscat	shine muscat	Fresh / Semifresh / Rotten	High-value grape variety
Strawberry	ttalgi	Fresh / Rotten	Fast spoilage

2. Methodology

2.1. Overall Pipeline

The classification pipeline consists of four sequential stages:

Stage	Component	Description
1	Data preparation	Raw datasets reorganized into unified `food/state/` folder structure
2	Feature extraction	Pre-trained CNN backbone (ImageNet weights), frozen or partially unfrozen
3	Projection head	$d_{\text{out}} \to 256 \to 128$ linear projection with ReLU
4	Classification	Linear layer $128 \to C$ with cross-entropy loss

All input images are resized to 224×224 pixels. Training images undergo data augmentation (random resized crop, horizontal flip, $\pm 30°$ rotation, color jitter on brightness/contrast/saturation/hue, Gaussian blur) to increase robustness to lighting and viewpoint variation. Validation images are center-cropped without augmentation. All images are normalized with ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

2.2. Backbone Architectures

Six backbone architectures are considered. Table 3 summarizes their properties. Preliminary experiments are conducted with ResNet-18 as the primary backbone. EfficientNet-B0 is identified as the recommended architecture for the final Korean food classifier based on its accuracy-to-parameter ratio and demonstrated suitability for small datasets.

Table 3: Backbone architectures considered in this study.

Model	Params	Out dim	ImageNet Top-1	Family
ResNet-18	11.2M	512	69.8%	ResNet
ResNet-34	21.3M	512	73.3%	ResNet
ResNet-50	25.6M	2048	76.1%	ResNet
MobileNetV3-S	2.5M	576	67.7%	MobileNet
EfficientNet-B0	5.3M	1280	77.1%	EfficientNet
EfficientNet-B2	9.1M	1408	80.1%	EfficientNet

2.3. Training Conditions

Four training conditions form an ablation study to determine the optimal degree of backbone fine-tuning. Conditions C3 and C4 (with projection head) are the focus of this study, as the additional non-linear projection consistently improves performance over a direct linear probe.

Table 4: Training conditions (ablation study).

ID	Name	Backbone	Head	Description
C1	`frozen`	Frozen	Linear only	No backbone adaptation
C2	`layer4`	Layer4 free	Linear only	Partial fine-tuning, no projection
C3	`head_frozen`	Frozen	Projection + Linear	Full head, frozen backbone
C4	`head_layer4`	Layer4 free	Projection + Linear	Full head + partial fine-tuning

All models are trained with Adam optimizer ($\mathrm{lr} = 10^{-3}$, $\mathrm{lr}_{\mathrm{backbone}} = 10^{-5}$, weight decay $= 10^{-4}$) and cosine annealing over 50 epochs. Batch size is 32.

2.4. Label Modes

Two classification targets are evaluated:

- **state**: classes correspond to freshness states only (fresh / rotten, or fresh / formalin / rotten). The model learns state-invariant features across all food categories.
- **fruit_state**: classes correspond to food-state combinations (e.g., `apple_fresh`, `apple_rotten`). The model simultaneously learns food identity and freshness state, enabling a richer confusion matrix analysis.

2.5. Evaluation Metrics

Given the potential class imbalance in real-world Korean food datasets, accuracy alone is insufficient. We report:

- **Macro F1-score**: harmonic mean of precision and recall, averaged equally across classes regardless of class frequency.
- **Macro Precision and Recall**: sensitivity to false positives and false negatives per class.
- **Confusion matrix**: row-normalized per true class, to identify systematic misclassification patterns.

For food safety applications, recall on the *rotten* class is particularly critical: a false negative (rotten food classified as fresh) carries higher risk than a false positive.

3. Preliminary Experiments

3.1. Setup

All experiments in this section use ResNet-18 with ImageNet pre-trained weights. Conditions C3 and C4 (with projection head) are the primary focus. A stratified 80/20 train/validation split is applied to all datasets. Experiments marked *pending* are in progress at the time of this report.

3.2. Results

Table 5 reports the best macro F1-score achieved per dataset and condition combination.

Table 5: Best macro F1-score (%) — ResNet-18, conditions C3 and C4. *Pending*: experiment in progress. N/A: fruit_state not applicable (single-fruit dataset). Bold: best result per dataset.

Dataset	C3: head_frozen		C4: head_layer4	
	state	fruit_state	state	fruit_state
FruitVision (3-class)	82.6	86.2	**88.0**	*pending*
Fruits Classification	94.0	*pending*	**95.8**	*pending*
Lemon Varieties	*pending*	N/A	**98.2**	N/A
Fruits Fresh/Rotten	*pending*	99.2	*pending*	**99.5**
Fresh & Stale (9 categories)	*pending*	*pending*	*pending*	*pending*
Fruits Quality	*pending*	*pending*	*pending*	*pending*

3.3. Confusion Matrix Analysis

Figure 1 shows the row-normalized confusion matrices for two representative completed experiments.

3.4. Learning Curves

Figures 2–5 show the per-epoch metrics for all completed runs, grouped by dataset. Each curve corresponds to one training condition and backbone combination.

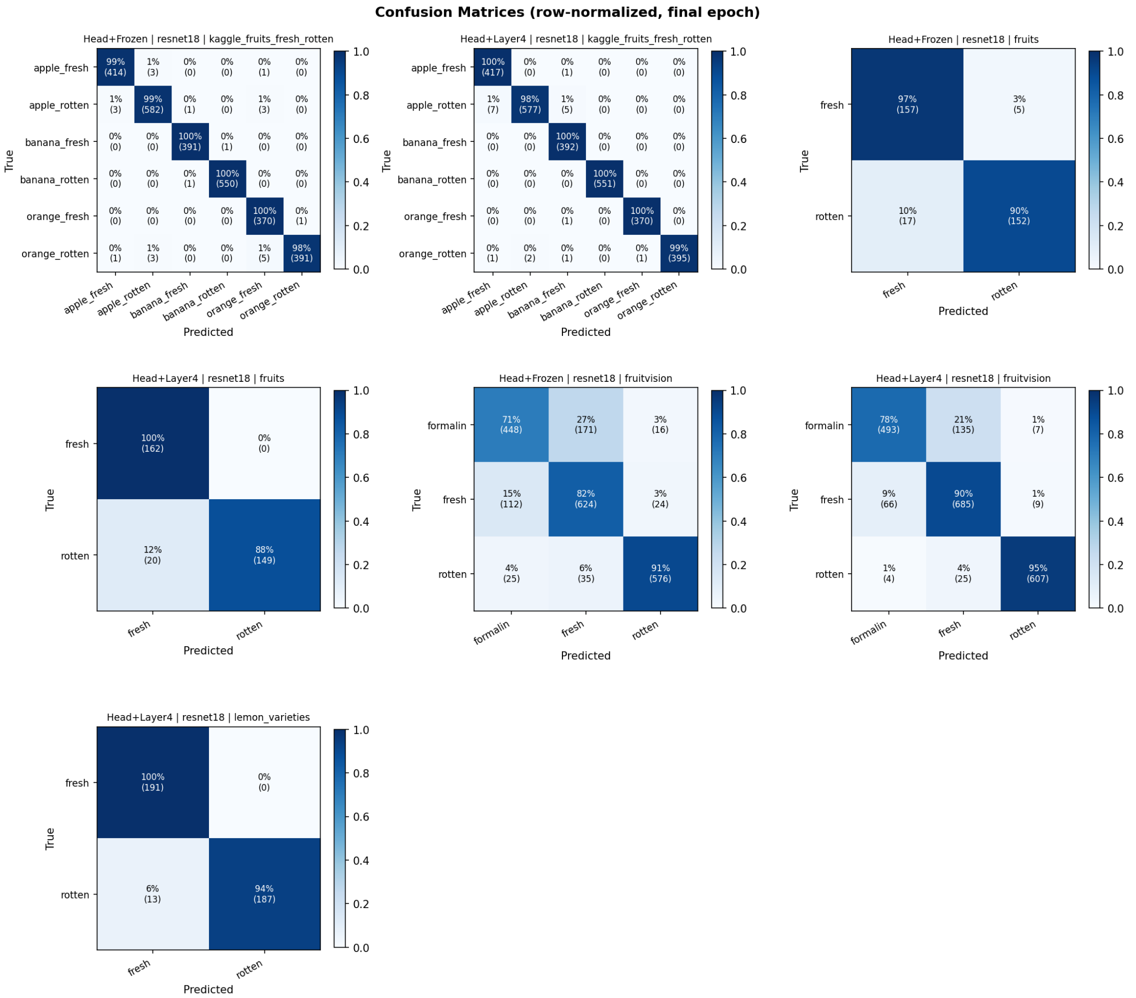

Figure 1: Row-normalized confusion matrices (final epoch). The formalin class in FruitVision shows the highest misclassification rate, predominantly confused with fresh — analogous to the challenge of detecting subtle spoilage in Korean fermented foods. The Kaggle Fruits Fresh/Rotten matrix (6 classes, fruit_state mode) shows near-perfect diagonal structure with essentially no cross-fruit confusion.

3.5. Key Findings

The following observations are drawn from completed experiments:

1. **Fine-tuning is critical for subtle spoilage detection.** On FruitVision (3-class), the gap between C3 (frozen backbone, 82.6%) and C4 (layer4 fine-tuned, 88.0%) is 5.4 percentage points. For binary problems, the gap is smaller but consistent: Fruits Classification achieves 94.0% (C3) vs. 95.8% (C4).

2. **Formalin is the hardest class to detect.** In FruitVision (C4, state mode), per-class F1 at 50 epochs: rotten 96.4%, fresh 85.4%, formalin 82.3%. The safety-critical error is formalin predicted as fresh: 135 out of 635 formalin samples (21.3%) pass undetected. In fruit_state mode (15 classes, C3), this dangerous error is reduced: only 9 out of 129 apple_formalin samples are predicted as apple_fresh (7.0%). However, fresh apples are frequently misclassified as formalin (69/153 = 45%), indicating the frozen backbone struggles to separate the two visually similar states and errs on the side of over-flagging.

Figure 2: Validation accuracy over training epochs per dataset.

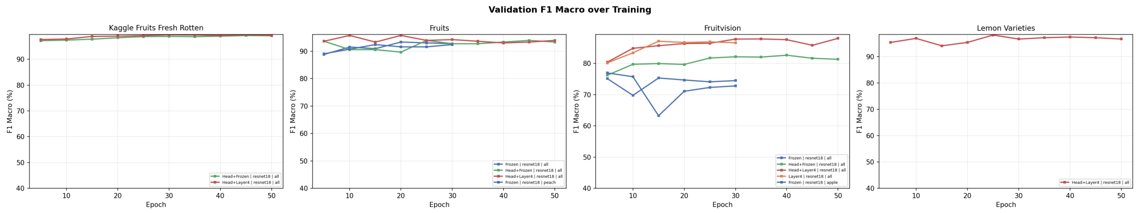

Figure 3: Macro-averaged F1 score over training epochs per dataset.

3. **fruit_state mode improves FruitVision with a frozen backbone.** C3 achieves 82.6% in state mode (3 classes) but 86.2% in fruit_state mode (15 classes). The additional fruit-identity signal helps the frozen backbone form better decision boundaries even without fine-tuning.
4. **Transfer learning reaches near-perfect accuracy on clean datasets.** Fruits Fresh/Rotten (apple, banana, orange) achieves 99.5% F1 with C4 in fruit_state mode (6 classes), with the confusion matrix showing near-zero off-diagonal entries — the model jointly learns fruit identity and freshness state without error.
5. **Small datasets converge early and require early stopping.** Lemon Varieties (1,956 images) peaks at epoch 25 (98.2% F1) and shows rising validation loss thereafter. Fruits Classification (1,655 images) peaks at epoch 10 (95.8% F1, C4) then oscillates. For the Korean food dataset, early stopping on validation F1 will be essential.

4. Korean Food Dataset Proposal

4.1. Target Categories and Classification Scheme

Two types of Korean food items are targeted:

Korean fruits, directly comparable to existing datasets: Korean apple, Jeju mandarin (*gyul*), shine muscat grape, Korean strawberry (*maehyang* variety). These share visual properties with Western counterparts but have distinct color profiles and surface textures that make a domain-specific model preferable.

Korean prepared foods, with no analog in existing datasets: Kimchi (fermented cabbage), *tteok* (rice cake), *banchan* (side dishes), cooked rice (*bap*). For these categories, binary classification (fresh / rotten) is proposed, as the intermediate "semifresh" state is visually ambiguous in fermented items.

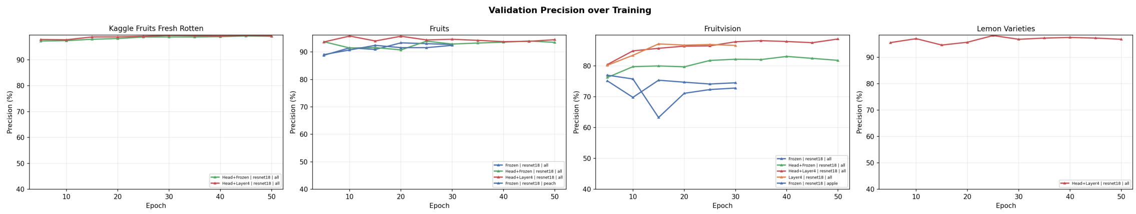

Figure 4: Macro-averaged precision over training epochs per dataset.

Figure 5: Macro-averaged recall over training epochs per dataset.

4.2. Data Collection Pipeline

Two automated image collection scripts have been implemented, one targeting Korean fruits and one targeting Korean prepared foods. Both use Bing Image Search with 5–8 search queries per class per freshness state, combining Korean-language and romanized search terms to maximize result diversity (e.g., *"fresh kimchi bright red color"* and *"moldy kimchi white fungus"* for the kimchi category).

Downloaded images are validated and processed as follows:

- **Integrity check**: corrupt or unreadable files are discarded.
- **Minimum resolution**: 100×100 pixels.
- **Format**: any format accepted; converted to RGB JPEG.
- **Output size**: 256×256 pixels (model crops to 224×224 during training).

The pipeline supports incremental collection with resume capability: if a class folder already contains images from a previous run, only the remaining count is downloaded. The target dataset size is 100 images per class per freshness state. Table 6 summarizes the expected dataset composition.

Table 6: Expected composition of the Korean food dataset (target counts).

Category	Type	States	Classes	Target image
Kimchi, tteok, banchan, bap	Prepared food	2 (Fresh/Rotten)	8	800
Apple, mandarin, strawberry	Korean fruit	2 (Fresh/Rotten)	6	600
Shine muscat, mandarin	Korean fruit	3 (Fresh/Semi/Rotten)	6	600
Total				**∼2,000**

This scale is deliberately comparable to Lemon Varieties (1,956 images), for which the pipeline achieved 98.2% F1 with transfer learning — confirming that 2,000 images is a sufficient starting point.

4.3. Integration with Training Pipeline

The Korean food dataset will be organized in the same `food/state/` nested structure used throughout this study, making it directly compatible with the existing training and evaluation pipeline without any code modifications. The `prepare_datasets.py` script can be extended with a new preparation function, and the training pipeline selects the dataset interactively at runtime.